

Sohrab Dhillon

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EDUCATION

University of Toronto

Sep. 2024 - April 2029

Bachelor of Applied Science and Engineering in Material Science Engineering + PEY Co-op

Toronto, ON

- Intended Minors: Advanced Manufacturing and Engineering Business
- Relevant Courses: Solid Mechanics, Thermodynamics, Heat and Mass Transfer, Diffusion & Kinetics, Engineering Statistics, Structures & Characterization of Materials

TECHNICAL SKILLS

CAD: SolidWorks, AutoCAD

Simulation: ANSYS FEA/CFD, Solidworks FEA

Additive Manufacturing: FDM, SLA, SLS, DfAM

EXPERIENCE

Structures Member

Sep. 2025 – Present

University of Toronto Aerospace Team UAS Structures Subteam

Toronto, ON

- Designed circular and oval lightening cutouts in SolidWorks using solid mechanics principles, reducing balsa airframe weight while limiting stress concentrations; incorporated plastic film covering to provide a smooth aerodynamic surface.
- Integrated balsa airframe structures, an SLA-printed nose cone, and locally reinforced base plates, tailoring material selection and geometry to balance low weight, aerodynamic shape, and structural support in regions subjected to high shear and concentrated loads.
- Prepared airframe geometry for laser cutting by transferring SolidWorks designs to AutoCAD and verifying scale, cutting layers, and export formats, preserving dimensional accuracy for fabrication and assembly.

Lifeguard & Swim Instructor

June 2026 – Sep 2026

City of Niagara Falls

Niagara Falls, ON

- Monitored high-traffic pool areas and taught swim lessons to groups of students, balancing safety supervision with individualized instruction during peak hours.
- Improved student swim proficiency by breaking down strokes into manageable steps, while conducting regular rescue drills and equipment checks to ensure a safe swimming environment.
- Responded to patron concerns and minor incidents by applying first aid and CPR training, ensuring quick resolution and upholding facility safety and service standards.

Simulation Lead

Dec. 2025 – Apr. 2026

Engineering Manufacturing & Design II, University of Toronto

Toronto, ON

- Evaluated four composite beam concepts using SolidWorks CAD, finite element analysis (FEA), and material trade-off studies; selected a corrugated balsa-CFRP design for its superior simulated stiffness over the XPS alternative and projected assembly material cost of \$3.74 against a \$7.00 budget.
- Engineered a modular coupling system for four beams using bending calculations, second-moment-of-area analysis, and FEA; developed a paired beam arrangement with ONYX slot-and-rod connectors that reduced predicted displacement and localized strain relative to the initial configuration.
- Refined laser-cut geometries in AutoCAD and verified connector fit during prototyping, applying design-for-manufacturing and assembly principles to standardize interfaces between balsa cores and 3D-printed ONYX rods, enabling repeatable alignment and tool-free assembly.

PROJECTS

Lightweight Composite Beam Design & Analysis

Sep. 2025 – Dec. 2025

Engineering Manufacturing & Design, University of Toronto

Toronto, ON

- Designed and evaluated composite beam concepts in SolidWorks with FEA-supported simulations, comparing stiffness, weight, material properties, and manufacturability to select a corrugated Balsa-CFRP beam design
- Performed structural analysis on beam coupling strategies, assessing bending stress and deformation, factor of safety, shear transfer, and material performance; implemented a slot-and-rod ONYX connector system

- Assisted with prototyping and manufacturing planning by optimizing laser-cut geometries using AutoCAD, verifying connector tolerances, and testing assemblies to ensure reliable load performance under a 0.5 kg midpoint load

Classroom Power Distribution System Design

Jan. 2025 – Apr. 2025

Engineering Strategies & Practice II, University of Toronto

Toronto, ON

- Designed a conceptual power distribution solution for an elementary school classroom, integrating four desk stations with 16 additional receptacles to improve access and reduce congestion at existing outlets.
- Evaluated 70+ concepts through feasibility screening, multivoting, and Pugh matrix analysis, narrowing alternatives to a preferred design based on client requirements and classroom constraints.
- Translated client needs into measurable design requirements and CAD layouts, incorporating a CDN \$250 budget, cable management, and workstation accessibility to guide concept development.
- Developed a verification plan referencing ISO 9241-210 for usability evaluation and ISO 14224 reliability principles, defining coverage, charging capacity, and maintenance criteria to assess the proposed design.

Sustainable Campus Food-Service System Design

Sep. 2024 – Dec. 2024

Engineering Strategies & Practice I, University of Toronto

Toronto, ON

- Translated site research and stakeholder needs into design criteria referencing CSA/ASC B651:23, Toronto Green Standard Version 4, and Codex food-labelling standards, establishing accessibility requirements and a 60% waste-diversion target.
- Collaborated with three teammates to screen 52 food-service proposals through multivoting and structured evaluation, producing a shortlist of three concepts aligned with client priorities.
- Compared shortlisted concepts against four criteria using a Pugh chart and impact-effort matrix, supporting the recommendation of a food-delivery robot and courtyard hub to address campus food accessibility.

CERTIFICATIONS

SOLIDWORKS Professional (CSWP) (in progress)

Oct. 2026 (expected)

Dassault Systèmes

SOLIDWORKS Associate (CSWA)

Jan. 2026

Dassault Systèmes

SOLIDWORKS Additive Manufacturing Associate (CSW-AM)

Nov. 2025

Dassault Systèmes

Lifeguard & Swim Instructor

Sep. 2025

Lifesaving Society Canada

Standard First Aid CPR-C

Aug. 2025

Lifesaving Society Canada

REFERENCES

Upon Request